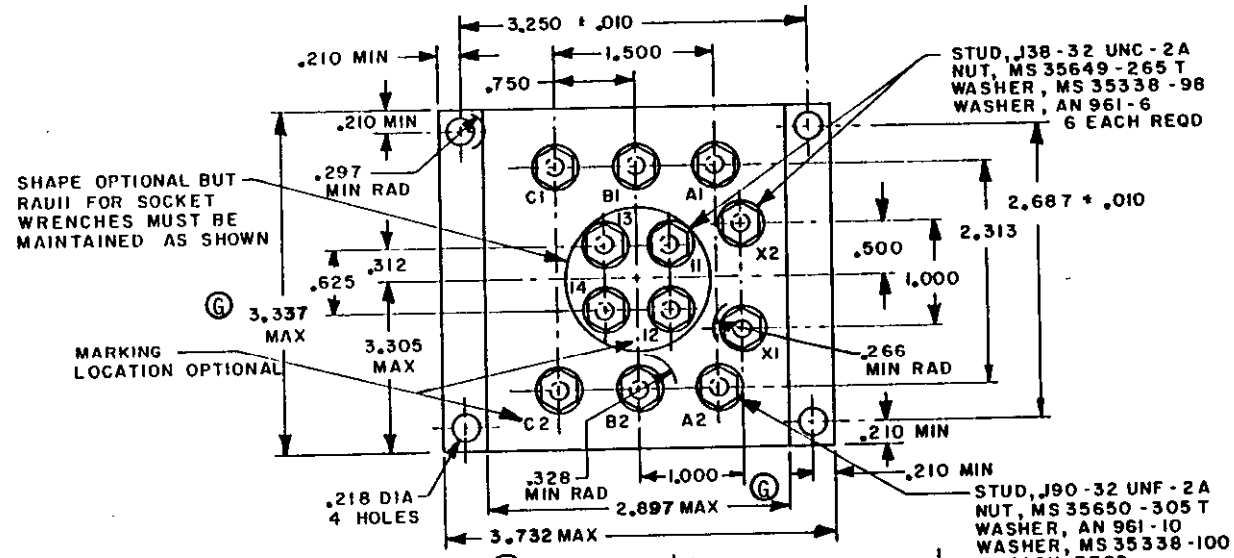
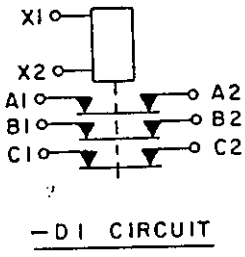
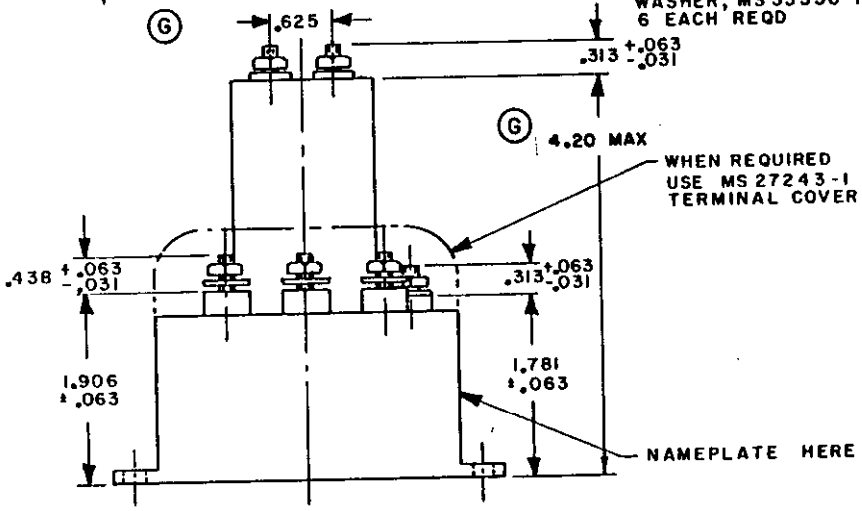
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5945

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NO SUPERSEDING STANDARD

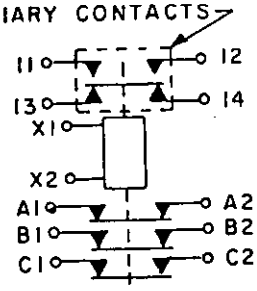



-D1 CIRCUIT




AUXILIARY CONTACTS




-D2 CIRCUIT

Ⓖ DENOTES CHANGES

APPROVED 14 Nov 1960 REVISED Ⓕ 5 JUN 87 Ⓖ 6 Jan 1989

User activities: Army - Navy - Air Force -
Review activities: Army - EC Navy - Air Force - 11, 99

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| | | | |
|---|---|---|---|
| P.A USAF - 85<br>Other Cust Navy - AS | International Interest | TITLE RELAYS, ELECTROMAGNETIC, 3 PST (NO), 12 AMPERES, TYPE I, HERMETICALLY SEALED | MILITARY STANDARD<br>MS21330 |
| Procurement Specification MIL-R-6106 | | SUPERSEDES: | PAGE 1 OF 5 |

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5945-0773-01

AMSC N/A

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .010 | 0.25 | .266 | 6.76 | .625 | 15.88 | 2.687 | 68.25 |
| .031 | 0.79 | .297 | 7.54 | .750 | 19.05 | 2.897 | 73.58 |
| .063 | 1.60 | .312 | 7.92 | 1.000 | 25.40 | 3.250 | 82.55 |
| .138 | 3.51 | .313 | 7.95 | 1.500 | 38.10 | 3.305 | 83.95 |
| .190 | 4.83 | .328 | 8.33 | 1.781 | 45.24 | 3.337 | 84.76 |
| .210 | 5.33 | .438 | 11.13 | 1.906 | 48.41 | 3.732 | 94.79 |
| .218 | 5.54 | .500 | 12.70 | 2.313 | 58.75 | 4.172 | 105.97 |
| | | | | | | 4.20 | 106.68 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.031 (0.79 mm).
4. Coil and auxiliary terminals may use additional washer for terminal seat.
5. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
6. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.

TABLE I. Dash numbers and characteristics.

| Dash number MS21330- | Type | Coil | Terminal type | Mounting or mating socket | Auxiliary contacts | Max weight in pounds |
|---|---|---|---|---|---|---|
| D1 | I | dc | Stud | Flange | None | 1.6 |
| D2 | I | dc | Stud | Flange | Yes | 1.6 |

User activities: Army - Navy - Air Force -
Review activities: Army - EC Navy - Air Force - 11, 99
APPROVED 14 Nov 1960 REVISED (G) FOR CHANGES SEE PAGES 1 AND 5
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| P.A | International Interest | TITLE | MILITARY STANDARD |
|---|---|---|---|
| USAF - 85<br>Other Cust<br>Navy - AS | | RELAYS, ELECTROMAGNETIC, 3 PST (NO), 12 AMPERES, TYPE I, HERMETICALLY SEALED | MS21330 |
| Procurement Specification<br>MIL-R-6106 | | SUPERSEDES: | PAGE 2 OF 5 |


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P.A
USAF - 85
Other Cust
Navy - AS

International Interest

Procurement Specification
MIL-R-6106

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 3 PST (NO), 12 AMPERES, TYPE I, HERMETICALLY SEALED

MILITARY STANDARD

MS21330



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TABLE II. Operating characteristics.

| MS21330-part or identifying number (PIN) | Coil data | | | | | | | | | | | Time (milliseconds maximum) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coil | Nominal | | | Max | | Max pick-up voltage | | | Hold voltage 2/ | Drop out voltage 2/ | Operate 3/ | Release 4/ | Contact bounce | | | |
| | | Volts 1/ | Freq Hz | Ω Res +15% -10% | Volts | Amperes | Normal 2/ | High temp test | Cont current test | | | | | Main | | Aux | |
| | | | | | | | | | | | | | | NO | NC | NO | NC |
| D1 | X1, X2 | 28 | dc | 51 | 29 | .6 | 18 | 21 | 22.5 | 7.0 | 1.5 | 25 | 10 | 2 | | | |
| D2 | X1, X2 | 28 | dc | 51 | 29 | .6 | 18 | 21 | 22.5 | 7.0 | 1.5 | 25 | 10 | 2 | | 7.5 | 7.5 |

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.

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Review activities: Army - Navy - EC Air Force - 11, 99

User activities: Army - Navy - Air Force -

TABLE III. Rated contact load (amperes per pole) case grounded.

| Type of load | Life operating cycles X $10^3$ | 28 V dc | | | | 115 V ac, 1 phase | | | | 115/200 V ac, 3 phase 1/ | | | | See appropriate notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | |
| Resistive | 200 2/ | 12 | | 1.25 | 1.25 | 12 | | 1.25 | | 12 | | | | |
| Inductive | 200 2/ | 12 | | .75 | .75 | 12 | | .75 | | 12 | | | | |
| Inductive | | | | | | | | | | | | | | |
| Motor | 200 | 6 | | | | 6 | | | | 6 | | | | |
| Lamp | | | | | | | | | | | | | | |
| Transfer load | | | | | | | | | | | | | | 3/ |
| Mechanical life Reduced current | 200 2/ | 3 | | .25 | .25 | 3 | | .25 | | 3 | | | | |
| Intmd current | 200 | 1.2 | | Per spec | | 1.2 | | | | 1.2 | | | | |

1/ Absence of value indicates relay is not rated for 3-phase applications.
2/ 100,000 for auxiliary contacts.
3/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

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Procurement Specification MIL-R-6106

P.A USAF - 85 Other Cust Navy - AS

International interest

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 3 PST (NO), 12 AMPERES, TYPE I, HERMETICALLY SEALED

MILITARY STANDARD MS21330

PAGE 4 OF 5

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TABLE IV. Vibration level.

| Dash no. MS21330- | 5 - 10 Hz | 10 - 55 Hz | 55 - 250 Hz | 250 - 500 Hz | 500 - 1500 Hz |
|---|---|---|---|---|---|
| D1<br>D2 | .08 DA | .06 DA | 10 g's | 5 g's | 4 g's |

TABLE V. Qualification by similarity.

| Dash no. MS21330- | Loads | | | | | | Dynamics 1/ | | | Environmental 2/ | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Type I | | | Type I ER | | | | | | | | |
| | A | B | C | D | E | F | A | X | Y | A | B | C |
| D1 | 4 | | | | | | 2 | | | 4 | | |
| D2 | 4 2/ | | | | | | 2 | | | 4 | | |

1/ All units must be tested, reference appendix I, MIL-R-6106, construction, internal.
2/ Test unit with auxiliary contacts, reference appendix I, MIL-R-6106, construction, internal.

## Environmental characteristics

| | |
|---|---|
| Temperature range | -70°C to +125°C |
| Max altitude rating | 80,000 ft |
| Shock G-level | 25 g's |
| Duration | 6-9 ms |
| Max duration contact opening | 2 ms |
| Vibration - sinusoidal (see table IV) | |
| Vibration - random | N/A |
| High shock Acceleration | 15 g's |

## Quality conformance inspection

Performance of groups B and C tests are not applicable.

(G) Group A acceptance reports shall be submitted to the qualifying activity on a yearly basis in order to retain qualification for this military standard sheet.

## Electrical characteristics

| | |
|---|---|
| Minimum insulation resistance, initial | 100 megohms |
| After life or environmental tests | 50 megohms |
| Dielectric strength (sea level) | 2-5 seconds |

| | Initial | | After life tests | |
|---|---|---|---|---|
| | 28 V dc | 115 V ac | 28 V | 115 V |
| Coil to case | 1,250 | 1,500 | 1,000 | 1,125 |
| Aux contacts | 1,250 | | 1,000 | |
| All other points | 1,800 | | 1,350 | |

| Dielectric strength (altitude) | 28 V dc | 1 minute 115 V ac |
|---|---|---|
| Coil to case | 500 | 500 |
| Aux contacts | 500 | 500 |
| All other points | 700 | 700 |

| | |
|---|---|
| Max contact drop initial | 0.150 volt |
| After life test | 0.175 volt |
| Overload current (NO) | 96 amperes |
| Rupture current (NO) | 120 amperes |
| Duty rating | Continuous |

User activities: Army - Navy - Air Force -
Review activities: Army - EC Navy - Air Force - 11, 99
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